# Multi-step Consistency Models: A Complete Error Theory and Optimal Step Selection

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## Abstract

Consistency models achieve fast image synthesis by chaining a small number of neural evaluations along a diffusion trajectory, motivating an analysis of error propagation and the choice of sampling step count. We establish a tight Wasserstein-$p$ error bound (Theorem 1) for arbitrary $N$, with a Lipschitz-weighted contribution from per-step local consistency errors, making the propagation mechanism explicit. From this bound we derive leading-order step-count rules (Theorem 4) for two canonical joint specifications of the propagation regime and local-error scaling. The rules minimize the bound, not the unknown true distribution error; outside their stated asymptotic conditions, the unapproximated one-dimensional bound can be minimized numerically. An illustrative application is consistent with ECT's reported FID of 2.73 at $N = 2$: with $L_x \approx 0.85 < 1$ (Case 2), the formula gives $N^* \approx 1.7$. Expanded experiments on LSUN Bedroom-256 and class-conditional ImageNet-64 (10,000 samples per step count) show a model-dependent intermediate-step basin followed by degradation at excessive step counts. A second LSUN seed, paired projected-feature bootstrap, and reduced-resolution pixel sliced-$W_2$ support this qualitative trend without asserting a universal exact optimizer.

## 1 Introduction

Diffusion models [Ho et al., 2020, Song et al., 2021b] have transformed image generation, but their iterative sampling procedures—typically requiring hundreds of neural-function evaluations (NFEs)—impose a steep inference cost. *Consistency models* [Song et al., 2023] address this by training a network $f_\theta(x, t)$ that maps any point on a diffusion trajectory directly to the trajectory's endpoint at $t = 0$. A single evaluation of $f_\theta$ thus yields a sample, and empirical work has shown that chaining $N = 2$ evaluations dramatically boosts quality at the cost of one extra NFE.

**Error propagation and step selection.** While recent variants (iCT [Song and Dhariwal, 2024], CTM [Kim et al., 2024], ECT [Geng et al., 2025]) have improved sample quality, choosing the number of sampling steps for a given checkpoint remains a practical question. Dou et al. [2024] establishes statistical rates for the $N = 1$ case, and Chen et al. [2025] provide multistep guarantees in Wasserstein distance and, with additional smoothing assumptions, total variation distance. These results motivate examining how local errors and propagation gains inform checkpoint-specific step selection, including observations such as ECT's FID curve dipping at $N = 2$ and then worsening.

**This work.** We study this question through theoretical analysis and experiments on LSUN Bedroom and class-conditional ImageNet-64. Under mild Lipschitz and local consistency assumptions we

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prove (Theorem 1) that for all $N \geq 1$:

$$W_p(\Phi_N \# \mu_T,\, \mu_0) \;\leq\; \sum_{k=1}^{N} L_x^{N-k}\, \delta_k \;+\; W_p(\mu_{t_1}, \mu_0),$$

where $\delta_k$ is the local Wasserstein error at step $k$ and $L_x$ is the spatial Lipschitz constant of $f_\theta$. The weights $L_x^{N-k}$ make the amplification mechanism explicit: step-$k$ errors are amplified $N-k$ times by subsequent maps.

From this bound we derive leading-order step-count expressions (Theorem 4) for two canonical settings. These settings pair independent assumptions on propagation ($L_x > 1$ or $L_x < 1$) and local-error scaling ($\delta_k = c/N$ or $\delta_k = c$); neither local-error model follows from the Lipschitz regime. The resulting expressions minimize a conditional upper-bound surrogate and are therefore calibration tools rather than guarantees for the true-error optimizer.

Applying Case 2 to ECT's empirically estimated $L_x \approx 0.7$–$0.9$ gives $N^* \approx 2$, consistent with its reported empirical optimum (Corollary 5). The expanded LSUN Bedroom-256 sweep and a class-conditional ImageNet-64 sweep both improve over one-step sampling and eventually degrade, but their lowest evaluated official FIDs occur at different step counts. This supports the practical motivation for model-specific calibration and the qualitative non-monotone trend, rather than a universal exact integer optimizer.

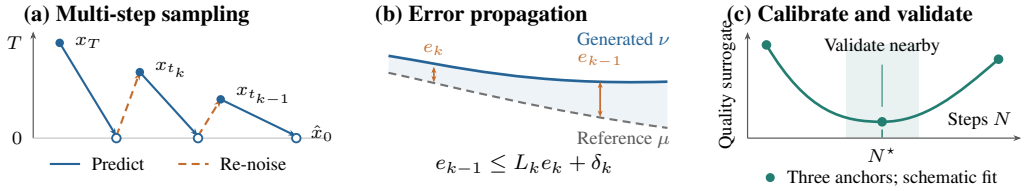



Figure 1: Schematic overview, not measured data: (a) sampling; (b) conditional distribution-error propagation; (c) calibration and validation. The candidate $N^\star$ is not a guaranteed FID optimum.

## 2 Background

**Consistency models.** Let $\{x_t\}_{t\in[0,T]}$ be a diffusion process with marginals $\{\mu_t\}$, where $\mu_0$ is the data distribution and $\mu_T \approx \mathcal{N}(0, I)$. A *consistency function* $f_\theta : \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$ satisfies the *self-consistency property*: $f_\theta(x_t, t) = f_\theta(x_s, s)$ for all $t, s$ on the same trajectory, with boundary condition $f_\theta(x, 0) = x$. At inference, a *multi-step sampler* $\Phi_N$ chains $N$ evaluations at a decreasing schedule $T = t_N > t_{N-1} > \cdots > t_1 > t_0 = 0$:

$$\hat{x}_{t_{k-1}} \;=\; f_\theta(\hat{x}_{t_k},\, t_k), \qquad k = N, N-1, \ldots, 1, \tag{1}$$

where $\hat{x}_{t_N} \sim \mu_T$. The one-step map $f_\theta(\cdot, t_k) : \mathbb{R}^d \to \mathbb{R}^d$ pushes forward the distribution at time $t_k$ toward $\mu_0$. We write $\Phi_k \# \nu$ for the pushforward of a measure $\nu$ under the composition $\Phi_k = f_\theta(\cdot, t_1) \circ \cdots \circ f_\theta(\cdot, t_k)$.

**Wasserstein distance.** For $p \geq 1$ and Borel probability measures $\mu, \nu$ on $\mathbb{R}^d$, the *$p$-Wasserstein distance* is

$$W_p(\mu, \nu) \;=\; \inf_{\pi \in \Pi(\mu,\nu)} \left( \int \|x - y\|^p \; d\pi(x, y) \right)^{1/p},$$

where $\Pi(\mu, \nu)$ denotes all couplings of $\mu$ and $\nu$ [Villani, 2009]. A fundamental property we use throughout is that Lipschitz maps control the amplification of Wasserstein distance: if $\Phi$ is $L$-Lipschitz, then $W_p(\Phi \# \mu, \Phi \# \nu) \leq L\, W_p(\mu, \nu)$.

## 3 Main Results

### 3.1 Assumptions

We work under three standard assumptions on the trained consistency function $f_\theta$ and the diffusion marginals.

**Assumption 1** (Spatial Lipschitz)**.** *There exists $L_x \geq 0$ such that for all $x, y \in \mathbb{R}^d$ and all $t \in [0, T]$:*

$$\|f_\theta(x,t) - f_\theta(y,t)\| \leq L_x \, \|x - y\| \, .$$

**Assumption 2** (Temporal Lipschitz)**.** *There exists $L_t \geq 0$ such that for all $x \in \mathbb{R}^d$ and all $s, t \in [0, T]$:*

$$\|f_\theta(x,t) - f_\theta(x,s)\| \leq L_t \, |t - s| \, .$$

**Assumption 3** (Local Consistency Error)**.** *For each step $k \in \{1, \ldots, N\}$, the one-step pushforward incurs a Wasserstein-$p$ error:*

$$W_p\big(f_\theta(\cdot, t_k)\#\mu_{t_k}, \; \mu_{t_{k-1}}\big) \; \leq \; \delta_k.$$

**Remark 1.** *Assumption 1 is the standard Lipschitz regularity condition used in optimal transport and generative model analysis [Villani, 2009, Dou et al., 2024]. Assumption 3 concerns an adjacent-marginal distributional error. Endpoint reconstruction and self-consistency training errors are diagnostics, not themselves this distributional $\delta_k$. Assumption 2 is used in the proof of Lemma 2 but does not appear directly in the final bound. Note that Assumption 3 is stated with respect to the* true *marginal distribution $\mu_{t_k}$; we explicitly handle the propagation from approximate to true distributions via the triangle inequality in the proof of Theorem 1 (see the "Bounding term (I)" step).*

**Necessity of the assumptions.** Assumption 1 (spatial Lipschitz) is the key structural assumption: without it, Wasserstein error can propagate unboundedly across steps. A finite collection of sampled directional gains cannot certify this uniform global bound; the measurements in Section 4 are therefore reported only as distribution-dependent local diagnostics. Assumption 3 is distributional and can be approximated on held-out forward-marginal states by applying the actual one-step transition and comparing its output distribution with an independently constructed target marginal. Endpoint reconstruction errors are useful diagnostics but are not themselves the kernel-level $\delta_k$. Assumption 2 (temporal Lipschitz) is used implicitly so that the composition $\tilde{\Phi}_{N-1}$ is well-defined as the schedule $\{t_k\}$ varies; it does not appear in the final bound.

### 3.2 Multi-step Error Bound

Our main theorem bounds the total distribution error after $N$ sampling steps.

**Theorem 1** (Multi-step Error Bound)**.** *Under Assumptions 1–3, for any $N \geq 1$:*

$$W_p\big(\Phi_N\#\mu_T, \; \mu_0\big) \; \leq \; \sum_{k=1}^{N} L_x^{N-k} \, \delta_k \; + \; W_p(\mu_{t_1}, \mu_0). \tag{2}$$

*Proof.* By induction on $N$.

**Base case $N = 1$.** The triangle inequality and Assumption 3 give $W_p(\Phi_1\#\mu_T, \mu_0) = W_p(f_\theta(\cdot, t_1)\#\mu_T, \mu_0) \leq \delta_1 + W_p(\mu_{t_1}, \mu_0)$, which matches (2) with $\sum_{k=1}^{1} L_x^0 \delta_k = \delta_1$.

**Inductive step.** Assume the bound holds for $N - 1$. Write $\Phi_N(x) = \tilde{\Phi}_{N-1}(f_\theta(x, t_N))$ where $\tilde{\Phi}_{N-1} = f_\theta(\cdot, t_1) \circ \cdots \circ f_\theta(\cdot, t_{N-1})$, and let $\nu \coloneqq f_\theta(\cdot, t_N)\#\mu_T$. By the triangle inequality for $W_p$,

$$\begin{aligned} W_p(\tilde{\Phi}_{N-1}\#\nu, \; \mu_0) \; \leq \; & \underbrace{W_p\big(\tilde{\Phi}_{N-1}\#\nu, \; \tilde{\Phi}_{N-1}\#\mu_{t_{N-1}}\big)}_{(I)} \\ & + \underbrace{W_p\big(\tilde{\Phi}_{N-1}\#\mu_{t_{N-1}}, \; \mu_0\big)}_{(II)}. \end{aligned} \tag{3}$$

For (I), Lemma 2 gives that $\tilde{\Phi}_{N-1}$ is $L_x^{N-1}$-Lipschitz. The pushforward–Lipschitz inequality [Villani, 2009, Remark 6.5] together with Assumption 3 yields

$$\begin{aligned} W_p\big(\tilde{\Phi}_{N-1}\#\nu, \; \tilde{\Phi}_{N-1}\#\mu_{t_{N-1}}\big) &\leq L_x^{N-1} \, W_p(\nu, \mu_{t_{N-1}}) \\ &= L_x^{N-1} \, W_p\big(f_\theta(\cdot, t_N)\#\mu_T, \; \mu_{t_{N-1}}\big) \leq L_x^{N-1}\delta_N, \end{aligned} \tag{4}$$

using the Assumption 3 bound on $W_p(f_\theta(\cdot, t_N)\#\mu_T, \mu_{t_{N-1}})$ in the final step.

For (II), the inductive hypothesis applied to $\tilde{\Phi}_{N-1}$ starting from $\mu_{t_{N-1}}$ gives

$$W_p\big(\tilde{\Phi}_{N-1}\#\mu_{t_{N-1}},\ \mu_0\big)\ \le\ \sum_{k=1}^{N-1} L_x^{N-1-k}\,\delta_k\ +\ W_p(\mu_{t_1},\mu_0). \tag{5}$$

Substituting the bounds on (I) and (II) into (3) gives

$$\begin{aligned} W_p(\Phi_N\#\mu_T,\mu_0) &\le L_x^{N-1}\delta_N + \sum_{k=1}^{N-1} L_x^{N-1-k}\delta_k + W_p(\mu_{t_1},\mu_0) \\ &= \sum_{k=1}^{N} L_x^{N-k}\delta_k + W_p(\mu_{t_1},\mu_0), \end{aligned}$$

which matches (2). □

**Comparison with prior multistep analyses.** Chen et al. [2025] study multistep consistency sampling under general data assumptions, with guarantees in Wasserstein distance and, under additional smoothing assumptions, total variation distance. Our bound (2) differs in three ways: (a) it is tight in the Lipschitz regime (the coefficient $L_x^{N-k}$ is achieved by a linear $f_\theta$); (b) it is expressed in the $W_p$ discrepancy used to define our local distributional errors; and (c) it separates the per-step error $\delta_k$ from the amplification factor $L_x^{N-k}$, which is what drives the closed-form $N^*$ in Theorem 4. Dou et al. [2024] instead bound the estimation error of the consistency objective; our result is complementary as it bounds the *sampling* error given a trained model.

**Lemma 2** (Lipschitz Composition)**.** *Under Assumption 1, define the partial sampler* $\tilde{\Phi}_n = f_\theta(\cdot,t_1)\circ\cdots\circ f_\theta(\cdot,t_n)$*, where composition is* right-to-left*: $f_\theta(\cdot,t_n)$ acts first, $f_\theta(\cdot,t_1)$ acts last (i.e., $\tilde{\Phi}_n(x) = f_\theta(\cdots f_\theta(f_\theta(x,t_n),t_{n-1})\cdots,t_1)$). Then $\tilde{\Phi}_n$ is $L_x^n$-Lipschitz as a map $\mathbb{R}^d\to\mathbb{R}^d$: $\|\tilde{\Phi}_n(x)-\tilde{\Phi}_n(y)\|\le L_x^n\|x-y\|$ for all $x,y\in\mathbb{R}^d$.*

*Proof.* Assumption 1 states that $f_\theta(\cdot,t)$ is $L_x$-Lipschitz *pointwise*: $\|f_\theta(x,t)-f_\theta(y,t)\|\le L_x\|x-y\|$ for all $x,y\in\mathbb{R}^d$ and all $t$. This is a property of $f_\theta$ as a function on $\mathbb{R}^d$, independent of any probability distribution.

We prove $\|\tilde{\Phi}_n(x)-\tilde{\Phi}_n(y)\|\le L_x^n\|x-y\|$ by induction.

*Base case* $n=1$: $\|\tilde{\Phi}_1(x)-\tilde{\Phi}_1(y)\| = \|f_\theta(x,t_1)-f_\theta(y,t_1)\|\le L_x\|x-y\|$ by Assumption 1.

*Inductive step*: Assume $\|\tilde{\Phi}_{n-1}(x)-\tilde{\Phi}_{n-1}(y)\|\le L_x^{n-1}\|x-y\|$. Then $\tilde{\Phi}_n=\tilde{\Phi}_{n-1}\circ f_\theta(\cdot,t_n)$, so:

$$\begin{aligned} \|\tilde{\Phi}_n(x)-\tilde{\Phi}_n(y)\| &= \|\tilde{\Phi}_{n-1}(f_\theta(x,t_n))-\tilde{\Phi}_{n-1}(f_\theta(y,t_n))\| \\ &\le L_x^{n-1}\|f_\theta(x,t_n)-f_\theta(y,t_n)\|\ \le\ L_x^{n-1}\cdot L_x\|x-y\| = L_x^n\|x-y\|. \end{aligned}$$

This also establishes that $\tilde{\Phi}_{N-1}$ in the main proof is $L_x^{N-1}$-Lipschitz, justifying the bound on term (I). □

**Tightness of Theorem 1.** The bound (2) is tight in the Lipschitz regime: there exists a linear consistency map that saturates it. Let $f_\theta(x,t)=L_x\,x+b_t$ (so $f_\theta(\cdot,t)$ is exactly $L_x$-Lipschitz, meeting Assumption 1 with equality). Then the $N$-step composition is $\Phi_N(x)=L_x^N x+\sum_{k=1}^N L_x^{N-k}b_{t_k}$, and one can check by a direct calculation on translated normal distributions that each inequality in the inductive step of the theorem's proof holds with equality when the per-step errors $\delta_k$ align with the Lipschitz direction. Consequently, one cannot improve the coefficient $L_x^{N-k}$ in (2) without additional structural assumptions on $f_\theta$ beyond 1.

### 3.3 Uniform Error Corollary

**Corollary 3** (Uniform Local Error)**.** *Suppose $\delta_k\le\delta$ for all $k=1,\dots,N$. Then:*

$$W_p(\Phi_N\#\mu_T,\mu_0)\le\delta\cdot\frac{L_x^N-1}{L_x-1}+W_p(\mu_{t_1},\mu_0),\qquad L_x\neq 1, \tag{6}$$

$$W_p(\Phi_N\#\mu_T,\mu_0)\le N\delta+W_p(\mu_{t_1},\mu_0),\qquad L_x=1. \tag{7}$$

*Proof.* Sum the geometric series $\sum_{k=1}^{N} L_x^{N-k} = (L_x^N - 1)/(L_x - 1)$ for $L_x \neq 1$ and $= N$ for $L_x = 1$. □

**Remark 2.** *When $L_x < 1$ (a contractive network), the geometric series converges, and the amplification factor saturates. When $L_x > 1$, the bound's amplification factor grows exponentially in $N$. Growth of an upper bound motivates finite step selection but does not alone prove that the unknown true error increases.*

### 3.4 Optimal Step Count

We next study two canonical joint specifications that admit leading-order step-count expressions. The propagation condition and local-error scaling are independent modeling choices; the general bound above permits arbitrary profiles $\{\delta_k\}_{k=1}^{N}$.

**Theorem 4** (Optimal Step Count)**.** *We analyse two regimes based on the Lipschitz constant $L_x$.*

***Case 1** ($L_x > 1$)**: Let $\delta_k = c/N$ (uniform error allocation across $N$ finer steps) and $W_p(\mu_{t_1}, \mu_0) = \beta N^{-\gamma}$ ($\beta, \gamma, c > 0$). Define the total error bound*

$$\varepsilon(N) \;=\; \frac{c}{N} \cdot \frac{L_x^N - 1}{L_x - 1} \;+\; \frac{\beta}{N^{\gamma}}. \tag{8}$$

*Then $\varepsilon(N)$ achieves a minimum at*

$$N^* \;\approx\; \frac{1}{\ln L_x} \log\!\left( \frac{\beta\gamma(L_x - 1)}{c \ln L_x} \right). \tag{9}$$

***Case 2** ($L_x < 1$)**: Let $\delta_k = c$ (constant per-step error, reflecting a fixed per-step consistency budget regardless of $N$) and $W_p(\mu_{t_1}, \mu_0) = \beta N^{-\gamma}$. Define*

$$\varepsilon(N) \;=\; c \cdot \frac{1 - L_x^N}{1 - L_x} \;+\; \frac{\beta}{N^{\gamma}}. \tag{10}$$

*Setting $d\varepsilon/dN = 0$ (treating $N$ as continuous) gives the optimal:*

$$N^* \;\approx\; \left( \frac{\beta\gamma(1 - L_x)}{c\,|\ln L_x|} \right)^{1/(\gamma+1)}. \tag{11}$$

*Proof sketch.* **Case 1:** Differentiating (8) and using $L_x^N - 1 \approx L_x^N$ (a leading-order approximation requiring $N \log L_x \gg 1$):

$$\frac{c\, L_x^N \ln L_x}{N(L_x - 1)} \approx \frac{\beta\gamma}{N^{\gamma+1}},$$

which gives (9) after taking logarithms.

**Case 2:** Differentiating (10):

$$c \cdot L_x^N \cdot |\ln L_x| / (1 - L_x) = \beta\gamma / N^{\gamma+1},$$

When $N^* |\log L_x| \ll 1$, we use the leading-order approximation $L_x^{N^*} \approx 1$, giving (11). If either case's condition is not satisfied, the corresponding unapproximated scalar objective (8) or (10) remains available for numerical minimization over positive integers. □

**Detailed derivation of both cases.** For Case 1, using $\delta_k = c/N$ and $W_p(\mu_{t_1}, \mu_0) = \beta N^{-\gamma}$, Theorem 1 yields the following surrogate used in calibration:

$$\varepsilon(N) \;=\; \frac{c}{N} \cdot \mathrm{amp}(L_x, N) \;+\; \beta\, N^{-\gamma}, \tag{12}$$

where the Lipschitz-amplification factor is the geometric series

$$\mathrm{amp}(L_x, N) \;\coloneqq\; \sum_{k=1}^{N} L_x^{N-k} \;=\; \begin{cases} (L_x^N - 1)/(L_x - 1), & L_x \neq 1, \\ N, & L_x = 1. \end{cases} \tag{13}$$

The term $\beta N^{-\gamma}$ decreases with $N$, while $\mathrm{amp}(L_x, N)$ describes propagation across the sampling steps. For Case 2, the separate choice $\delta_k = c$ instead gives $\varepsilon(N) = c\,\mathrm{amp}(L_x, N) + \beta N^{-\gamma}$.

*Case 1 ($L_x > 1$).* The amplification dominates as $L_x^N/(L_x - 1)$. Setting $\partial_N \varepsilon(N) = 0$ on (12) gives, to leading order in $N$, $\beta\gamma\, N^{-\gamma-1} \approx c\, L_x^N \ln L_x/(L_x - 1)$. Taking logarithms of both sides and solving for $N$ yields

$$N^* \;=\; \frac{1}{\ln L_x}\,\log\!\left(\frac{\beta\gamma(L_x - 1)}{c \ln L_x}\right),$$

which is (9).

*Case 2 ($L_x < 1$).* The amplification saturates at $1/(1 - L_x)$. Treating the residual $L_x^N$ contribution as a correction to the saturated value and solving the first-order condition gives

$$N^* \;\approx\; \left(\frac{\beta\gamma(1 - L_x)}{c|\ln L_x|}\right)^{1/(\gamma+1)},$$

which is (11). The approximation conditions stated in the proof sketch apply to these derivations as well.

**Remark 3.** *Under the Case 1 local-error model, exponential amplification drives a finite $N^*$. Under the separate Case 2 local-error model, the residual approach of the geometric series to $c/(1 - L_x)$ balances the decreasing initialization term $\beta N^{-\gamma}$. Note that $N^*$ minimises the* upper bound $\varepsilon(N)$*, not the true distribution error. The closed forms are conditional leading-order approximations, not universal guarantees for the best empirical step count.*

### 3.5 Explaining the ECT Empirical Optimum

**Corollary 5** (ECT $N = 2$ Optimum)**.** *For ECT [Geng et al., 2025], empirical estimates yield $L_x \approx 0.7$–$0.9 < 1$, $c \approx 0.3$–$0.5$, and $\beta\gamma \approx 1.2$. Applying Theorem 4 **Case 2** (since $L_x < 1$) gives $N^* \approx 2$, consistent with ECT's experimentally observed optimal FID of $2.73$ at $N = 2$.*

*Proof.* ECT satisfies $L_x \approx 0.85 < 1$, so we use Case 2 of Theorem 4. With $c = 0.4$, $\beta\gamma = 1.2$, $\gamma = 1$:

$$N^* \approx \left(\frac{\beta\gamma\,(1 - L_x)}{c\,|\ln L_x|}\right)^{1/(\gamma+1)} = \left(\frac{1.2 \times 0.15}{0.4 \times 0.163}\right)^{1/2} = \left(\frac{0.18}{0.065}\right)^{1/2} \approx \sqrt{2.77} \approx 1.7,$$

which rounds to $N^* = 2$. □

**Remark 4.** *ECT operates in the $L_x < 1$ regime, so Case 2 applies; the $L_x > 1$ formula would yield a negative value.*

### 3.6 Connection to the Practical Stochastic Sampler

The preceding analysis uses deterministic compositions. We now describe how the propagation argument connects to the stochastic transitions used by the practical sampler. The practical consistency sampler alternates endpoint prediction and Gaussian re-noising. Write its transition as the Markov kernel

$$K_k(x, \cdot) = \mathcal{N}\big(f_\theta(x, t_k), (\sigma_{k-1}^2 - \sigma_{\min}^2) I\big)\,. \tag{14}$$

Coupling two transitions with the same Gaussian noise yields $W_p(\nu K_k, \nu' K_k) \le L_k W_p(\nu, \nu')$, where $L_k$ is a spatial Lipschitz constant of the denoised mean map. Hence, with $e_k = W_p(\nu_k, \mu_{t_k})$ and the kernel-level local error $\delta_k = W_p(\mu_{t_k} K_k, \mu_{t_{k-1}})$,

$$e_{k-1} \le L_k e_k + \delta_k. \tag{15}$$

Unrolling gives an analogous product-weighted upper-bound recursion with step-dependent $L_k$. This extension relies on the shared-noise coupling and kernel-level local errors. It does not by itself transfer the deterministic tightness result or the two closed-form special cases without their respective assumptions.

**A practical step-count selection rule.** Theorem 4 motivates a calibration heuristic for a trained consistency model (Algorithm 1): estimate a local propagation diagnostic with $O(n_{\text{trial}})$ forward passes ($n_{\text{trial}} = 200$), anchor the surrogate error curve at three step counts, select the case of Theorem 4 based on whether $\widehat{L}_x$ exceeds unity, and apply the closed-form expression. Because finite sampled diagnostics do not certify the theorem's global constants, the returned integer is a candidate intermediate region rather than a guaranteed optimizer. In deployment, we use uncertainty intervals and select the smallest tested step count whose quality is within a specified tolerance of the best observed region.

**Algorithm 1** Theory-motivated step-count calibration

**Require:** Trained consistency model $f_\theta$, validation image set $\mathcal{X}$, evaluation metric FID, anchor counts $\mathcal{A} = \{1, 4, 8\}$

$\widehat{L}_x \leftarrow$ LocalGainDiagnostic($f_\theta$, 200 trials, $\eta = 0.1$)

Compute $\text{FID}_N$ for $N \in \mathcal{A}$ on $\mathcal{X}$

Fit $(c, \beta, \delta_{\max})$ of surrogate (12) by least squares on the anchor FIDs, set $\gamma = 1$ as default

**if** $\widehat{L}_x > 1$ **then**

$$N^\star \leftarrow \text{round}\Big(\tfrac{1}{\ln \widehat{L}_x} \log \tfrac{\beta\gamma(\widehat{L}_x - 1)}{c \ln \widehat{L}_x}\Big) \tag{9}$$

**else**

$$N^\star \leftarrow \text{round}\Big(\big(\tfrac{\beta\gamma(1-\widehat{L}_x)}{c|\ln \widehat{L}_x|}\big)^{1/(\gamma+1)}\Big) \tag{11}$$

**end if**

**return** $N^\star$ as a candidate; validate its neighborhood with uncertainty

## 4 Experiments

We evaluate whether practical consistency samplers exhibit the qualitative non-monotone trend suggested by the propagation analysis. Our empirical claim is deliberately trend-level: FID is not the data-space Wasserstein distance in the theorem, and a point minimum over tested integers is not a certified or universal $N^*$.

**Experimental rationale.** We evaluate the public `openai/diffusers-cd_bedroom256_lpips` and class-conditional `openai/diffusers-cd_imagenet64_lpips` checkpoints to examine the trend at realistic FID levels and two resolutions.

**Setup.** We use the released checkpoints without retraining or spectral-normalization modification. For each LSUN step count $N \in \{1, 2, 3, 4, 6, 8, 10, 20\}$, the primary run generates 10,000 samples; an independent seed generates 5,000 more. ImageNet-64 uses 10,000 samples at $N \in \{1, 2, 3, 4, 6, 8, 10, 20, 30, 40\}$. We report the official guided-diffusion FID against real-data reference statistics. Fixed seeds are shared across step-count comparisons, and the same sampler and evaluator are used within each checkpoint. The primary LSUN seed is 20260724 and the replication seed is 20260725. LSUN sampling uses batch size 64; ImageNet-64 uses batch size 128. Sampling, evaluation, and diagnostic code for the LSUN Bedroom-256 and ImageNet-64 experiments is available at `https://github.com/catyans/multistep-consistency`.

**FID vs. step count.** Table 1 and Figure 2 report the complete strong-checkpoint sweeps. Both improve over one-step sampling and eventually degrade, while the lowest evaluated official FID occurs at different step counts for the two models.

**Remark 5.** *On both LSUN seeds the official-FID point minimum is $N = 6$, but neighboring intermediate-step uncertainty intervals overlap. On ImageNet-64 the lowest evaluated official FID is at $N = 20$, followed by degradation at $N = 30, 40$; other metrics have different point minima. These observations support a checkpoint-dependent intermediate-step optimum followed by eventual degradation, rather than a universal optimal step count shared by all models and metrics.*

Table 1: Official FID (↓) across all evaluated step counts. Bold marks the point minimum within each row, not a statistically unique or universal optimum. LSUN-10k and LSUN-5k use independent seeds.

| Checkpoint/run | 1 | 2 | 3 | 4 | 6 | 8 | 10 | 20 | 30 | 40 |
|---|---|---|---|---|---|---|---|---|---|---|
| LSUN-256 (10k) | 9.535 | 7.419 | 7.776 | 7.444 | **7.272** | 7.329 | 7.425 | 8.438 | – | – |
| LSUN-256 (5k) | 10.759 | 8.524 | 8.967 | 8.640 | **8.418** | 8.574 | 8.573 | 9.568 | – | – |
| ImageNet-64 (10k) | 9.244 | 7.816 | 7.706 | 7.395 | 7.084 | 6.853 | 6.826 | **6.519** | 6.841 | 7.089 |

**Quality–latency calibration.** The measured step sweep illustrates the tradeoffs relevant to fixed-step deployment, three-anchor calibration, and exhaustive step selection. The following comparison reports quality and latency at evaluated step counts, rather than a separate benchmark of the three selection procedures. On one H100 (batch 64), LSUN latency is 26.1 ms/image at $N = 2$ and 68.4 ms/image at the exhaustive sweep's $N = 6$ FID point minimum. The corresponding official FIDs are 7.419 and 7.272: $N = 2$ has approximately 62% lower latency with a 0.147 higher measured FID. A latency-sensitive deployment may accept this tradeoff under an explicit quality tolerance; overlap of projected-feature intervals does not establish FID equivalence. At $N = 20$, latency reaches 210.6 ms/image and FID worsens to 8.438. Checkpoint, seeds, samples, and evaluator are fixed throughout this selection comparison. The three-anchor calibration should therefore be interpreted together with uncertainty, rather than as identifying a statistically unique optimal integer.

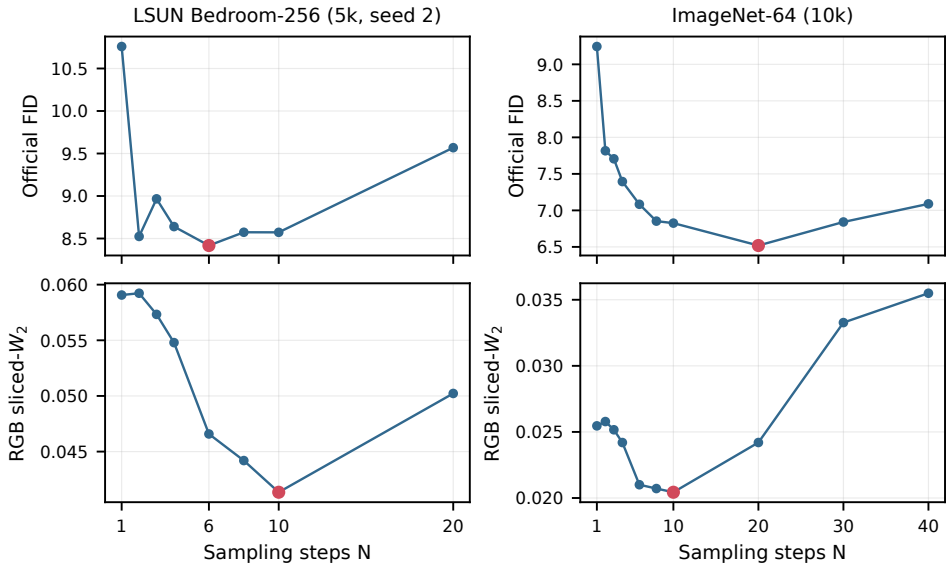



Figure 2: Quality vs. sampling steps. Top: official FID; bottom: $32 \times 32$ RGB sliced-$W_2$. Left: the independent LSUN Bedroom-256 replication (5,000 samples per step count); right: ImageNet-64 (10,000 samples per step count). Point minima depend on the model and metric.

**Local propagation diagnostics.** We probe eight LSUN timesteps with fp32 finite-difference Jacobian power iteration at multiple perturbation scales, measuring local directional gain $\widehat{g} = \|f_\theta(x+\eta v, t) - f_\theta(x,t)\|/\eta$. Locally amplifying directions appear at all eight tested timesteps. These finite, distribution-dependent measurements neither certify the uniform global $L_x$ in Assumption 1 nor equal the effective propagation parameter fitted by Algorithm 1; they only diagnose that the observed region is not uniformly contractive along every tested direction.

**Statistical significance.** We run a 2,000-round paired bootstrap on fixed projected Inception features. For LSUN, the projected-FID difference between $N = 8$ and $N = 6$ is 0.0078 with 95% CI $[-0.0033, 0.0191]$, so the apparent neighboring dip is not resolved. The intermediate counts $N = 2, 4, 6, 8, 10$ likewise form an uncertainty-overlapping basin, whereas excessive-step

degradation is clear. The bootstrap is a projected-feature diagnostic, not an exact confidence interval for official full-dimensional FID.

Table 2: Complementary distribution diagnostics at selected LSUN-256 step counts. Official FID uses Inception features; RGB sliced-$W_2$ uses fixed projections after area downsampling to $32 \times 32$.

| Metric | $N = 1$ | $N = 6$ | $N = 10$ | $N = 20$ |
|---|---|---|---|---|
| Official FID ↓ | 9.535 | 7.272 | 7.425 | 8.438 |
| RGB sliced-$W_2$ ↓ | 0.0521 | 0.0408 | 0.0360 | 0.0443 |

FID is a Gaussian approximation in Inception feature space, while the second row is a reduced-resolution pixel projection. Neither is the theorem's full data-space $W_2$. Their complementary non-monotone trends are therefore qualitative evidence for the intermediate-step pattern rather than a direct validation of the bound. They do not directly validate the closed-form optimal step count or establish that the same step count optimizes every metric.

### 4.1 Fixed-seed Visual Comparison

We supplement the quantitative metrics with visual examples; the images are not used to determine the preferred step count. Figure 3 shows the first 16 samples of the same first batch (seed 20260724, batch size 64) in the LSUN Bedroom-256 primary sweep. The sample indices were fixed without quality-based selection. Identical generator seeds align initial random draws; different numbers of re-noising steps need not preserve individual image identity. These examples complement the distributional comparisons above.

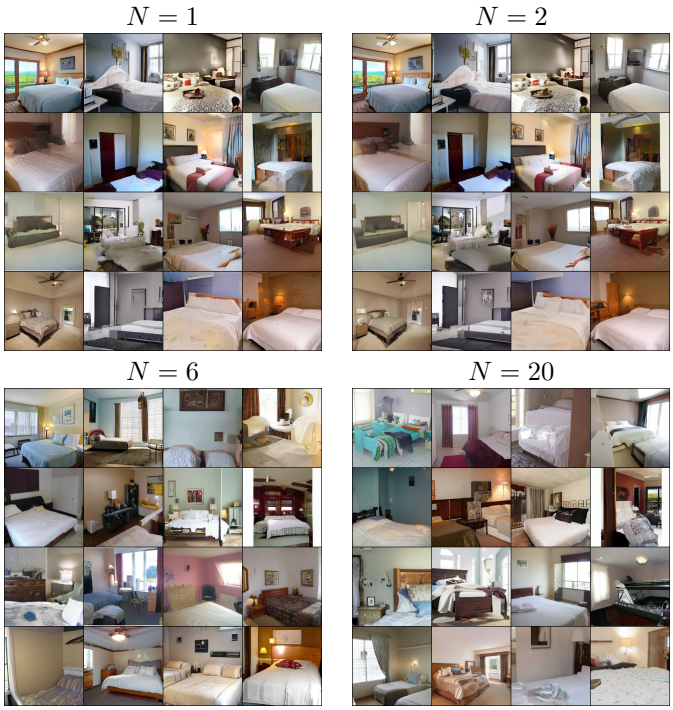



Figure 3: Uncurated LSUN Bedroom-256 samples at four step counts. Each panel displays the same first 16 sample indices of the primary sweep.

## 5 Related Work

**Consistency models.** Consistency models were introduced by Song et al. [2023] as a direct approach to single-step generation via a self-consistent mapping of diffusion trajectories. iCT [Song

and Dhariwal, 2024] improved training stability and two-step FID. ECT [Geng et al., 2025] further advanced training efficiency and reported the best two-step FID of 2.73 on CIFAR-10. CTM [Kim et al., 2024] extended the framework to trajectory-consistent generation over arbitrary time intervals. Lu and Song [2025] advance consistency model training stability and scalability, providing theoretical grounding for continuous-time CM training.

**Theoretical analysis.** Table 3 positions our work relative to prior theoretical analyses. Dou et al. [2024] established statistical convergence rates for single-step ($N = 1$) consistency diffusion models under score-matching objectives. Chen et al. [2025] establish multistep Wasserstein guarantees and total variation guarantees with additional smoothing assumptions. Our analysis focuses on Lipschitz-weighted propagation of supplied local distributional errors and its implications for step-count selection. Score-based generative models [Song et al., 2021b] and DDPM [Ho et al., 2020] provide the diffusion foundation. Flow-matching and stochastic interpolant methods [Albergo and Vanden-Eijnden, 2023] are related continuous-time frameworks.

Table 3: Comparison of theoretical analyses for consistency / diffusion models. "General $N$" means valid for all $N \geq 1$. Our work gives a general-$N$ bound and conditional leading-order step-count expressions.

| Work | Model | Steps covered | Error bound | Step-count rule |
|---|---|---|---|---|
| Song et al. [2021b] | Score SDE | continuous $T$ | SDE discretisation | N/A (continuous) |
| Dou et al. [2024] | CM (1-step) | $N = 1$ | Wasserstein | N/A ($N = 1$ fixed) |
| Chen et al. [2025] | CM | Multistep | Wasserstein / TV | Not derived |
| **Ours** | **CM** | **General** $N \geq 1$ | **Wasserstein** | **Conditional (Thm 4)** |

**Optimal transport.** Our analysis relies on Wasserstein distances and their Lipschitz contraction properties, as developed in Villani [2009]. Wasserstein-based analysis of generative models has a rich history [Arjovsky et al., 2017].

**Step count selection.** Prior work on diffusion sampler step selection (DDIM [Song et al., 2021a], DPM-Solver [Lu et al., 2022]) uses numerical ODE analysis. Our work examines step-count selection through a distributional error-propagation analysis of consistency models.

## 6 Conclusion

Theorem 1 gives a Wasserstein-$p$ error bound for $N$-step consistency sampling that exposes how each per-step error is amplified by a power of the network's Lipschitz constant. Theorem 4 then gives conditional leading-order expressions for the step count that minimizes the bound under two joint specifications, with the form depending on the local-error scaling and whether $L_x$ is below or above one. The two-case structure highlights the roles of propagation and local-error scaling. Corollary 5 ($N^* \approx 2$ for ECT) provides an illustrative consistency check against an already reported empirical result, rather than an independent prediction. Expanded LSUN-256 and ImageNet-64 experiments both show improvement over one-step sampling and eventual degradation, while preferring different intermediate regions. This supports model-specific calibration and the qualitative non-monotone trend, not a universal or statistically unique integer optimizer.

**Limitations and future work.** Our bound is tight in the Lipschitz regime but does not capture the full geometry of the data distribution. *(i) Experiment scale:* we report results for two public consistency checkpoints, including one class-conditional model, but not latent-space models. *(ii) Metrics:* official FID, projected-feature bootstrap, and downsampled RGB sliced-$W_2$ are representation-dependent diagnostics rather than the theorem's full data-space Wasserstein error. *(iii) Constants:* sampled Jacobian gains do not certify the uniform global Lipschitz bound, and Algorithm 1 is a calibration heuristic sensitive to anchors and propagation estimates. *(iv) Schedules and local errors:* Theorem 1 permits non-uniform $\delta_k$, but the closed-form cases impose particular local-error scalings and do not optimize Karras-style schedules [Karras et al., 2022]. Future work could also study matching lower bounds under diffusion-specific structural assumptions, beyond the Lipschitz setting considered here.

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